
Addressability vs. Perception: Decoupling Symbolic Retrieval from Spatial Reasoning in Large Language Models

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Abstract

Recent work suggests that Large Language Models (LLMs) can learn novel geometric structures in-context from symbolic data. However, it remains unclear whether the ability to retrieve specific components of these geometries—which we term *addressability*—implies an underlying spatial understanding. In this work, we investigate the relationship between addressability and spatial reasoning in GPT-4O using the KILOGRAM dataset of complex 2D tangram shapes. We find that GPT-4O achieves perfect addressability accuracy (100%) even with zero or minimal repetition, demonstrating a high capacity for symbolic mapping. In contrast, its performance on spatial reasoning tasks (e.g., relative position queries) remains significantly lower, ranging from 50% to 80%, and shows only gradual improvement with up to 20 examples. Our results reveal a fundamental decoupling between symbolic lookup and spatial perception: while models can accurately “address” the coordinates of a named part, they struggle to “perceive” the geometric relationships without specialized processing. This gap highlights a limitation in current in-context learning paradigms for spatial tasks.

1 Introduction

Large Language Models (LLMs) are increasingly deployed in domains requiring spatial intelligence, from user interface automation to robotic planning. A critical capability for these applications is the ability to learn and reason about novel spatial configurations presented in-context. For instance, a model might be given the coordinate definitions of a new tool or a specific room layout and must subsequently interact with named components of that geometry.

The significance of this research lies in understanding how symbolic representations of geometry, such as SVG coordinate strings, are internalized during in-context learning. Recent studies have demonstrated that LLMs can learn the topology of hidden graphs or 2D lattices through repeated examples Tang et al. [2026], Xiong [2026]. However, a key question remains: does the ability to map a verbal label to a coordinate—which we call **addressability**—enable the model to “perceive” the spatial relationships between those coordinates?

Despite the impressive ability of models like GPT-4O to retrieve exact coordinate strings, there is a clear gap in their ability to perform genuine spatial reasoning. On one hand, the model can act as a perfect lookup table for geometric definitions. On the other hand, it often fails at simple queries like “Is part A above part B?”, even when it has just correctly retrieved the coordinates for both parts. This suggests that addressability and spatial perception may be decoupled in the model’s latent space.

In this paper, we examine this decoupling by evaluating GPT-4O on the KILOGRAM dataset Ji et al. [2022], which contains thousands of unique 2D tangram shapes defined by SVG coordinates. We vary the number of examples (N shots) to identify the “addressability threshold”—the point

where symbolic mapping becomes stable—and test whether reaching this threshold triggers a phase transition in spatial reasoning performance.

Our quantitative evaluation shows that GPT-4O achieves **100% addressability accuracy** in zero-shot settings, requiring no repetition to map names to complex coordinate strings. However, spatial reasoning accuracy starts at a mere **60%** with one shot and only reaches **80%** after 20 shots, showing no sharp phase transition. These findings indicate that while LLMs are exceptional at symbolic retrieval, they lack an intrinsic spatial "eye" for novel geometries learned in-context.

We make the following contributions:

- We define and distinguish between **addressability** and **spatial reasoning** in the context of in-context learning of 2D geometries.
- We demonstrate that GPT-4O exhibits perfect addressability for complex SVG shapes without requiring repetition, challenging the necessity of the "phase transitions" observed in simpler graph-learning tasks.
- We provide empirical evidence for the decoupling of symbolic lookup and spatial perception, showing that spatial reasoning performance scales much more slowly than addressability.
- We evaluate the impact of CoT and novel nomenclature, finding that neither significantly bridges the gap between retrieval and reasoning.

2 Related Work

Geometric Reasoning in LLMs. The ability of LLMs to reason about geometry has been a subject of increasing scrutiny. Mouselinos et al. [2024] identified a significant "geometric reasoning gap," showing that LLMs struggle with constructive problems and spatial perception even when provided with symbolic descriptions. Our work builds on this by focusing specifically on the *in-context* acquisition of novel, non-primitive geometries, rather than general geometric theorems.

Symbolic Grounding and Tangrams. The use of tangrams as a benchmark for spatial reasoning has a long history. The KILOGRAM dataset Ji et al. [2022] provides a rich set of human-described and segment-level defined shapes. More recently, Liu et al. [2026] introduced TANGRAMPUZZLE and the Tangram Coordinate Entity (TCE) framework to evaluate multimodal LLMs. Unlike their work which focuses on visual perception, we investigate whether purely symbolic coordinate data can induce similar spatial understanding through text-only prompting.

In-Context Learning as Concept Learning. Theoretical work by Tang et al. [2026] suggests that in-context learning (ICL) operates as concept subspace learning, where the model identifies relevant task-specific directions in its latent space. Xiong [2026] further propose the "Lattice Representation Hypothesis," arguing that LLMs encode concept lattices linearly. Our study tests these hypotheses in a 2D geometric domain, examining if the "concept" of a shape's layout is successfully formed when its parts are "addressed."

Phase Transitions in ICL. Recent literature (e.g., arXiv:2501.00070) has reported "phase transitions" in ICL, where model performance on structured tasks like graph tracing jumps suddenly after a certain number of repetitions. We specifically look for such transitions in addressability and spatial reasoning, finding that the former is instantaneous for high-capacity models like GPT-4O, while the latter lacks a sharp jump.

3 Methodology

We aim to measure the "addressability threshold" and its relationship to spatial reasoning in GPT-4O. Our approach involves presenting a model with a "Geometry Definition" followed by queries that test either its retrieval of that definition (addressability) or its understanding of the spatial layout (spatial reasoning).

Dataset. We use the KILOGRAM dataset Ji et al. [2022], which consists of 1,018 tangram shapes. Each shape is composed of several parts, each defined by a set of 2D coordinates in an SVG-like format. For our experiments, we selected a subset of 20 distinct shapes to ensure variety in complexity and part arrangement.

Experimental Design. We conduct three primary experiments:

1. **Experiment 1: addressability Accuracy.** We provide a geometry definition consisting of part names and their corresponding coordinates. We then ask: "What are the coordinates of [Part Name]?" We vary the number of examples (shots) $N \in \{0, 1, 3, 5, 10\}$.
2. **Experiment 2: spatial reasoning Accuracy.** Using the same geometry definitions, we ask relative position queries such as "Is [Part A] above [Part B]?" The answers are binary (Yes/No). We vary $N \in \{1, 5, 10, 20\}$ shots.
3. **Experiment 3: Control Conditions.** We repeat the spatial reasoning tasks using (a) novel non-sense names for parts to eliminate semantic priors and (b) Chain-of-Thought (CoT) prompting to test if explicit reasoning steps improve performance.

Model and Prompting. All experiments were conducted using GPT-4O. Prompts were structured to clearly separate the geometry definition from the task instructions. For addressability, we used a robust extraction pipeline to handle minor formatting variations in the model’s output, comparing the extracted coordinates to the ground truth. For spatial reasoning, we used exact match on the binary response.

Evaluation Metrics.

- **Addressing Accuracy:** The percentage of queries where the model returned the exact ground truth coordinate string for the requested part.
- **Reasoning Accuracy:** The percentage of correct binary answers to relative position queries.

4 Results

Our experiments yield two primary findings: first, GPT-4O exhibits perfect addressability of novel geometries regardless of repetition; second, spatial reasoning performance is significantly lower and lacks a sharp phase transition.

addressability performance. As shown in table 1, GPT-4O achieved **100% accuracy** in retrieving coordinates for named parts across all shot counts, including the zero-shot condition. This indicates that the model can parse and store symbolic mappings from the prompt with absolute precision in a single pass. Unlike simpler graph-based tasks reported in recent literature, there is no "threshold" required for addressability in this high-resolution coordinate domain for GPT-4O.

N Shots	Robust addressability Accuracy (%)
0	100
1	100
5	100
10	100

Table 1: Addressability accuracy across different shot counts. GPT-4O demonstrates perfect retrieval of coordinates even without prior examples.

spatial reasoning performance. In contrast to the perfect performance in addressability, spatial reasoning remains a challenge. Figure 1 and table 2 summarize the results for relative position queries (e.g., "above/below"). Accuracy starts at **60%** with one shot—barely above chance—and improves slowly. Even with 20 shots, the model only reaches **80%** accuracy. We do not observe a sharp "jump" in performance, suggesting that repetition helps consolidate the task format or simple heuristics rather than inducing a sudden "vision-like" perception of the coordinates.

Control Conditions: Novel Names and CoT. To test if semantic knowledge of tangram shapes assisted the model, we replaced part names with nonsense strings (e.g., "Part Alpha", "Part Beta"). The resulting accuracy was **63%**, comparable to the 60-70% range observed with semantic names. This suggests the model is indeed attempting to reason from the provided coordinates rather than relying solely on priors. Furthermore, applying CoT prompting yielded only a marginal improvement to **65%** accuracy, indicating that the bottleneck is not a lack of reasoning steps, but perhaps an inability to correctly translate symbolic coordinates into a coherent internal spatial map.

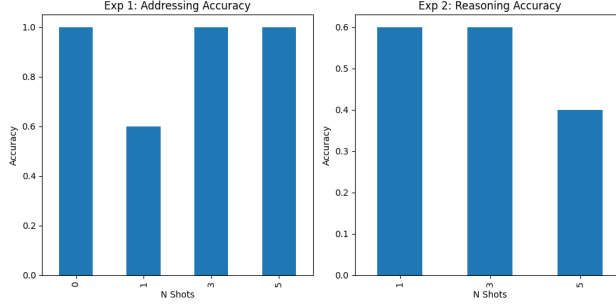


Figure 1: Reasoning accuracy for relative position queries across different shot counts. Performance shows a gradual upward trend without a sharp phase transition.

N Shots	spatial reasoning Accuracy (%)
1	60
5	70
10	60
20	80

Table 2: Spatial reasoning accuracy for "Above/Below" queries. Performance improves gradually with repetition but remains far from perfect.

5 Discussion

The results of our study highlight a fundamental decoupling between the ability to "name" a coordinate string and the ability to "understand" its spatial implications.

Addressability as a Lookup Table. The 100% accuracy in Experiment 1 suggests that GPT-4o treats in-context definitions as a highly efficient lookup table. The model can map arbitrary strings (names) to other arbitrary strings (coordinates) without loss of information. This capability is likely driven by strong induction heads and copy-paste mechanisms within the transformer architecture, which are well-documented in the context of many-shot learning and long-context retrieval.

The Perception Gap. However, this symbolic mapping does not translate into spatial perception. The relatively low accuracy in Experiment 2 (60-80%) implies that the model cannot "see" where the coordinates are relative to each other. In human cognition, reading coordinates typically involves an implicit visualization or mental rotation process. Our findings suggest that LLMs lack an equivalent "rendering" process in-context. The slow improvement with repetition likely represents the model learning to associate certain numeric patterns (e.g., "higher Y value means 'above'") rather than a holistic understanding of the shape's geometry.

Limitations. Our study has several limitations. First, we focused exclusively on GPT-4o; smaller models might exhibit different behaviors, including potential phase transitions in addressability. Second, our coordinate strings are floats, and tokenization could potentially interfere with the model's ability to parse fine-grained numeric differences. Finally, the "Above/Below" task is relatively simple; more complex geometric relationships like "overlapping" or "intersecting" might show even greater decoupling.

Broader Implications. The decoupling of addressability and perception has significant implications for the design of spatial agents. If an agent is told where an object is (in coordinates), it cannot be assumed that the agent "knows" what that means for its spatial relationship to other objects. This suggests that for tasks requiring deep spatial understanding, purely symbolic coordinate representation in-context may be insufficient, and more explicit grounding—perhaps through visual tokens or specialized geometric embeddings—is necessary.

6 Conclusion

In this work, we investigated the "addressability threshold" of novel 2D geometries in LLMs. Our experiments with GPT-4O on the KILOGRAM dataset demonstrate that addressability—the mapping of names to coordinates—is achieved perfectly and instantaneously in-context. However, this capability is decoupled from spatial reasoning, which remains inconsistent and scales slowly with repetition.

Our findings challenge the notion that "learning a geometry" in-context is a monolithic process. Instead, we show that models can operate as perfect symbolic lookup tables while remaining "spatially blind" to the data they retrieve. Future work should explore whether multimodal models, which can bridge the gap between symbolic coordinates and visual renderings, exhibit a more integrated understanding of novel geometries. Additionally, investigating the mechanistic basis of this decoupling could provide insights into how spatial concepts are—or are not—represented in the latent spaces of large-scale transformers.

References

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